Exercise sheet 1 – Solution

Linear Optimization (WS 26/27)

Classroom task 1.1

Consider the matrix

$$A = \begin{pmatrix} 0 & 1 & 0 \\ 2 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}.$$

How many ways do you know to characterize invertibility of A? Which one is the most useful to see or prove invertibility of A?

Some equivalent characterizations for a square 3×3 matrix are:

- A has an inverse matrix A^{-1}.
- $\det(A) \neq 0$.
- $\operatorname{rank}(A) = 3$.
- The rows of A are linearly independent.
- The columns of A are linearly independent.
- The homogeneous system $Ax = 0$ has only the solution $x = 0$.
- For every $b \in \mathbb{R}^3$, the system $Ax = b$ has exactly one solution.
- The columns of A span $\mathbb{R}^3$.
- 0 is not an eigenvalue of A.

For the given matrix, the determinant test is useful:

$$\begin{aligned} \det(A) &= 0 \cdot 0 \cdot 0 + 1 \cdot 1 \cdot 1 + 0 \cdot 2 \cdot 1 - 0 \cdot 0 \cdot 1 - 1 \cdot 2 \cdot 0 - 0 \cdot 1 \cdot 1 && \text{Sarrus rule} \\ &= 0 + 1 + 0 - 0 - 0 - 0 && \text{multiply all factors} \\ &= 1 && \text{add and subtract the terms} \end{aligned}$$

Check by expansion along the first row:

$$\begin{aligned} \det(A) &= -\det \begin{pmatrix} 2 & 1 \\ 1 & 0 \end{pmatrix} && \text{only the entry } a_{12} = 1 \text{ contributes} \\ &= -\big(2 \cdot 0 - 1 \cdot 1\big) && \text{determinant of a } 2 \times 2 \text{ matrix} \\ &= -(0 - 1) && 2 \cdot 0 = 0 \text{ and } 1 \cdot 1 = 1 \\ &= 1 && 0 - 1 = -1 \text{ and } -(-1) = 1 \end{aligned}$$

$$\boxed{\det(A) = 1 \neq 0, \text{ so } A \text{ is invertible.}}$$

Exercise 1.1

(1) Ask an AI of your choice the following question: "What is a linear program in standard form?" Compare the answer to the definition of standard form in the lecture notes. (2) Continue by asking the AI about each of the definitions of feasible or infeasible vector, feasible set, feasible or infeasible linear program, optimal value, bounded or unbounded linear program and optimal solution. Do the answers match the definitions in the lecture notes? For the submission, copy or write down the key parts of the answers of the AI and just tick it, if it is the same definition and if not, comment on the differences.

(1)

Key part of the AI answer:

$$\begin{array}{ll} \text{maximize} & c^T x \\ \text{subject to} & Ax = b, \\ & x \geq 0, \end{array}$$

where $x \in \mathbb{R}^n$ is the decision variable, $c \in \mathbb{R}^n$, $A \in \mathbb{R}^{m \times n}$ and $b \in \mathbb{R}^m$ are given data.

Comparison with the lecture notes:

This matches the lecture definition of standard form. ✓

Comment: If an AI answer also allows minimization problems or inequality constraints in the displayed standard form, then it is more general than the lecture definition. In the lecture notes, these forms are transformed into the standard form above by changing signs, adding slack variables, and replacing free variables by the difference of two nonnegative variables.

(2)

The key definitions from the AI answer and the comparison with the lecture notes are:

Term	Key part of the AI answer	Same as lecture notes?
feasible vector	A vector $x \in \mathbb{R}^n$ is feasible if it satisfies all constraints $Ax = b$ and $x \geq 0$; otherwise it is infeasible.	✓
feasible set	The feasible set is $C = \{x \in \mathbb{R}^n \mid Ax = b,\ x \geq 0\}$.	✓
feasible linear program	A linear program is feasible if $C \neq \emptyset$.	✓
infeasible linear program	A linear program is infeasible if $C = \emptyset$.	✓
optimal value	The optimal value is $v = \sup\{c^T x \mid x \in C\}$; for an infeasible LP the convention is $v = -\infty$.	✓
unbounded linear program	A feasible LP is unbounded if for every $M \in \mathbb{R}$ there is an $x \in C$ with $c^T x > M$; then $v = +\infty$.	✓
bounded linear program	A feasible LP is bounded if there is an $M \in \mathbb{R}$ such that $c^T x \leq M$ for every $x \in C$.	✓
optimal solution	A feasible vector $x^* \in C$ is an optimal solution if $c^T x^* \geq c^T x$ for every $x \in C$.	✓

All listed definitions match the lecture notes.

Exercise 1.2

A student has a total of 30 minutes to figure out the course requirements. The sources have the following values and times: asking a colleague: value 4*, time* 5 *min; coming to the first lecture: value* 8*, time* 9 *min; reading the description in the Moodle course: value* 7*, time* 7 *min; reading the LV-Karte: value* 16*, time* 15 *min; doing nothing to not risk the headache: value* 40*, time* 30 *min. Since we do not want to rely only on one source, we need to use at least two. The goal is to maximize the personal value within the time constraints. Formulate the optimization problem.*

Let

$$x_i = \begin{cases} 1, & \text{if source } i \text{ is used,} \\ 0, & \text{if source } i \text{ is not used,} \end{cases} \qquad i = 1, \ldots, 5.$$

The indices correspond to the five sources in the order stated in the exercise.

The objective coefficients are the values:

$$(4, 8, 7, 16, 40).$$

The time coefficients are the required minutes:

$$(5, 9, 7, 15, 30).$$

The time limit is 30 minutes and at least two sources must be used. Hence the model is

$$\begin{array}{ll} \text{maximize} & 4x_1 + 8x_2 + 7x_3 + 16x_4 + 40x_5 \\ \text{subject to} & 5x_1 + 9x_2 + 7x_3 + 15x_4 + 30x_5 \leq 30, \\ & x_1 + x_2 + x_3 + x_4 + x_5 \geq 2, \\ & x_i \in \{0, 1\}, \qquad i = 1, \ldots, 5. \end{array}$$

Check of the modelling choices:

- The first constraint counts minutes and compares them with 30.
- The second constraint counts the selected sources and requires at least 2.
- The binary condition means that every source is either selected once or not selected.

The optimization problem is the binary linear program displayed above.

Exercise 1.3

The company "FreshShine" produces three products. Product 1, Citrus Shower Gel, yields a profit of 3 *euros per unit. Product 2, Citrus Shampoo, yields a profit of* 5 *euros per unit. Product 3, Citrus Body Lotion, yields a profit of* 4 *euros per unit. Production requires machines A, B and C. Shower Gel needs* 1 *hour on A and* 1 *hour on B. Shampoo needs* 2 *hours on A,* 1 *hour on B and* 3 *hours on C. Body Lotion needs* 1 *hour on A,* 2 *hours on B and* 1 *hour on C. The available capacities are* 170 *hours on A,* 150 *hours on B and* 180 *hours on C. The company can sell all units produced. The goal is to maximize profit. Formulate the mathematical optimization model.*

Let

$$\begin{aligned} x_1 &= \text{number of units of Citrus Shower Gel produced per month,} \\ x_2 &= \text{number of units of Citrus Shampoo produced per month,} \\ x_3 &= \text{number of units of Citrus Body Lotion produced per month.} \end{aligned}$$

Profit per month is

$$3x_1 + 5x_2 + 4x_3.$$

Machine A is used for

$$1x_1 + 2x_2 + 1x_3$$

hours. Machine B is used for

$$1x_1 + 1x_2 + 2x_3$$

hours. Machine C is used for

$$0x_1 + 3x_2 + 1x_3$$

hours.

Therefore the linear optimization model is

$$\begin{array}{ll} \text{maximize} & 3x_1 + 5x_2 + 4x_3 \\ \text{subject to} & x_1 + 2x_2 + x_3 \leq 170, \\ & x_1 + x_2 + 2x_3 \leq 150, \\ & 3x_2 + x_3 \leq 180, \\ & x_1, x_2, x_3 \geq 0. \end{array}$$

Check of the modelling choices:

- The coefficient row $(1, 2, 1)$ is the monthly use of machine A per unit of products $1, 2, 3$.
- The coefficient row $(1, 1, 2)$ is the monthly use of machine B per unit of products $1, 2, 3$.
- The coefficient row $(0, 3, 1)$ is the monthly use of machine C per unit of products $1, 2, 3$.
- The nonnegativity constraints exclude negative production quantities.

The mathematical optimization model is the LP displayed above.